Monitoring of Wheat Powdery Mildew Disease Severity Using Multiangle Hyperspectral Remote Sensing

Li He, Shuang-Li Qi, Jian-Zhao Duan, Tian-Cai Guo, Wei Feng, and De-Xian He

***Abstract*— The pathogenesis of wheat powdery mildew (WPM) is from the bottom to upper layers of the plant, and the vertical observation angle limits the early monitoring of WPM status. Multiangle remote sensing could effectively extract spatial structural information from different plant layers. The objectives of this study were to improve the monitoring accuracy of WPM severity and to screen suitable observation angles by developing a novel vegetation index. The monitoring capacities of the novel parameters [Normalized Powdery Mildew Index (NPMI) and Ratio Powdery Mildew Index (RPMI)] and 14 optimized traditional spectral parameters were compared at 13 observation angles and different angle ranges. The correlation between all spectral parameters and disease severity was superior in the forward observation direction than in the backward observation direction in the principal plane of the Sun, and the correlation between the two observation directions decreased with an increase in observation angle. The new spectral parameter suitable for inversion of disease index was RPMI ($R_{744}/R_{762} - 0.5 \times R_{710}$), and the best optimal observation angle was +10°, with a 0.852 coefficient, which was 31.74% higher than that of the two-band spectral parameter, R_{744}/R_{762}. The fitting accuracy of the new parameter in the range of 0° to +30° in the forward direction was 0.704. RPMI could not only improve the monitoring accuracy of powdery mildew severity at a single angle but also achieve a more stable monitoring accuracy in the 0° to +30° range in the forward direction, which significantly expands the application scope of remote sensing monitoring and enhances the flexibility of the technology in actual production environments.**

***Index Terms*— Band combination, disease severity, observation angle, powdery mildew, wheat.**

I. Introduction

WHEAT powdery mildew (WPM) is a typical airborne disease threatening wheat production in China. With the promotion of dwarf and semidwarf wheat varieties, increasing planting density, irrigation, and fertilizer application, the WPM range is expanding, and the damage associated with the disease is becoming more severe. The occurrence of diseases and insect pests often leads to a decrease in plant biomass, the destruction of leaf structures, and a decrease in leaf chlorophyll and water concentrations [1]. A change in biochemical components would inevitably translate into different absorption and reflection characteristics which would be evident in the spectral reflectance of the plant [2], which provide a theoretical basis and technical support for the analysis and real-time monitoring of powdery mildew using rapid and nondestructive hyperspectral remote sensing technologies.

Numerous studies have reported that the sensitive bands screened by remote sensing monitoring for different types of plant species and plant diseases are different [3]. The reflectance change in the 450–850 nm range was associated with glume blight disease based on rice ear spectroscopy analyses [4]. The sensitive bands of rice leaf folder focused on 503–521, 526–545, 550–568, 581–606, 688–699, 703–715, and 722–770 nm at the leaf level and 747–754 nm at the canopy level [5]. The sensitive wavebands of WPM were concentrated in the red band (600–630 nm) and red edge (690–718 nm) at the canopy scale [6]; however, on the leaf scale, 400–500 and 610–690 nm positively responded to WPM disease [7]. Conversely, the wavebands sensitive to powdery mildew in sugar beet leaves with the disease severities from 1% to 30% were reported to be 450, 720, and 500–600 nm when infected leaves were collected until three weeks after inoculation [3]. The extensive exploration of sensitive wavebands for different types of diseases could facilitate the improvement of the accuracy of disease monitoring.

Based on sensitive wavebands, some scholars have screened and constructed vegetation indices (VIs), which further combine and enhance the original waveband information, and have applied it in modeling analyses of disease and insect pests in remote sensing monitoring. The red-edge slope and red-edge area were screened for the monitoring of WPM at the canopy scale [8], while 520, 584, and 724 nm were selected for the development of a powdery mildew index (PMI), which could better identify the powdery mildew in sugar beet [3]. Double green normalized difference index composed of green bands (584 nm, 550 nm) at the canopy scale, which could be used to estimate the severity of WPM in the middle and late growth stages [9]. Wavelengths including 417, 539, and 668 nm were

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selected to construct Fusarium classification index (FCI) for *Fusarium* head blight detection [10]. The canopy spectral data of multitemporal wheat stripe rust under water and fertilizer stress conditions were compared and Physiological Reflectance Index (PhRI) could be used to distinguish wheat stripe rust and water stress to some extent [11]. Since different diseases present different symptoms, the specificity indices used to characterize various diseases are different. Even in the case of the same disease, the indicator spectral parameters are different under different crop growth states.

To determine remote sensing data comprehensively for the monitoring of plant diseases and pests, a variety of algorithms or their combinations have been studied. Wavelet features and Fisher's linear discriminant analysis (FLDA) have been evaluated for the monitoring of yellow rust and WPM diseases, respectively [12], [13], and combined continuous wavelet analysis (CWA) and partial least squares (PLSR) have been explored for the estimation of powdery mildew severity on leaf level [14]. In addition, combined mixture tuned matched filtering (MTMF) and PLSR could improve the monitoring accuracy of WPM [15]. Meanwhile, CWA could be used to discriminate yellow rust, powdery mildew, and aphid [16], and these three common diseases and insect-associated damage can also be discriminated using FLDA and PLSR [17], in addition to VIs-based kernel discriminant approach [18]. Factor analysis and the back propagation neural network were also combined to achieve the accurate monitoring of WPM at different stages of the disease [19], while deep convolutional neural network (DCNN) could recognize anthracnose, downy mildew, powdery mildew, and target leaf spots of cucumber under field conditions [20].

The above studies demonstrate that the application of hyperspectral remote sensing technologies could facilitate the monitoring of the occurrence of plant diseases in real time. However, remote sensing monitoring of all diseases is carried out based on a vertical observation angle, and the actual occurrence characteristics of disease are not considered. The incidence of powdery mildew is from the bottom plant layer to the upper plant layer. In addition, the disease spots are more widespread in the front leaves than in the back leaves, and the disease information is mostly concentrated in the middle and lower parts of the canopy. As the disease progresses, the bacteria spread to the upper layers of wheat plants. The vertical observation direction mainly collects growth data from the upper part of the canopy, and the middle and lower layers of canopy growth data have a minor contribution. Therefore, data on early disease development may not be obtained in time. However, multiangle remote sensing technology can collect comprehensive data on WPM early lesion development on the upper, middle, and lower layers of the canopy based on different observation directions; therefore, it could improve the monitoring accuracy of disease and play a key role in early warning systems, disease prevention, and control of WPM.

Scholars have previously applied multiangle remote sensing data to invert physiological parameters or determine the appropriate observation angles for target parameters. The relationship between multiangle canopy spectral data and leaf nitrogen content (LNC) was analyzed comprehensively, and multiangle vegetation index was developed to improve the estimation accuracy of wheat LNC [21]. Other studies have also demonstrated that the correlation between spectral information and photosynthetic efficiency [22] and Leaf Area Index (LAI) [23] at nonperpendicular observation angles is more accurate than the correlation at vertical angles. Hot and dark point information is often introduced into traditional VIs in multiangle remote sensing to enhance the LAI monitoring accuracy [24]. Multiangle remote sensing is applied not only to improve the accuracy of monitoring target parameters but also to screen the sensitive observation angles corresponding to the parameters. For example, (0°, ±20°, and ±30°), (0°, ±30°, and ±40°), and (0°, ±50°, and ±60°) could separately invert the chlorophyll concentrations of the lower, middle, and upper layers [25], respectively. A combination of three observation angles (0°, 36°, and 55°) could accurately invert LAI and revealed that the inversion accuracy of LAI decreased when the number of observation angles applied exceeded three [26]. A multiangle observation approach has been applied to collect soybean canopy map data at different growth stages in the field and observed that the 40° view zenith angle (VZA) was the optimal angle for inverting chlorophyll density [27]. In addition, the LNC of winter wheat could be estimated stably in the backward observation angle range (−10° to −40°) [28]. Similarly, the saturation factor (SF) has been developed, which facilitates Optimized of Soil Adjusted Vegetation Index (OSAVI) × SF to establish a unified modeling model based on 13 observation angles [29], which expands the application angle of the sensor considerably.

Previous studies on diseases have mainly focused on vertical observation, while multiangle remote sensing studies have mainly focused on abiotic stress conditions. The spectral parameters proposed by previous studies under abiotic stress conditions may not be suitable for monitoring powdery mildew, and the spectral parameters developed at vertical angles may not be suitable for multiangle monitoring of powdery mildew. Therefore, how to apply multiangle remote sensing in monitoring WPM disease status, particularly, early symptoms from the lower layers of plant requires further study, which would facilitate early prevention and precise control of disease. The objectives of this study are as follows: 1) to develop a novel multiangular spectral parameter for monitoring the severity of WPM disease; 2) to compare the performance of traditional VIs and the novel spectral parameters at multiple angles; and 3) to determine the appropriate observation angle and remote sensing angle range for monitoring powdery mildew. The results of this study could facilitate the accurate monitoring of disease and the appropriate treatment of winter WPM, in addition to providing a theoretical basis and technical support for aerial remote sensing large-scale monitoring of wheat diseases.

II. Materials and Methods

A. Experimental Design

Experiment 1 (Exp.1): The inoculation test was conducted in a field at the Henan Agricultural University Science and Education Demonstration Park (34° 51′ N, 113° 35′ E) in

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Fig. 1. Development stages of the infected winter wheat.

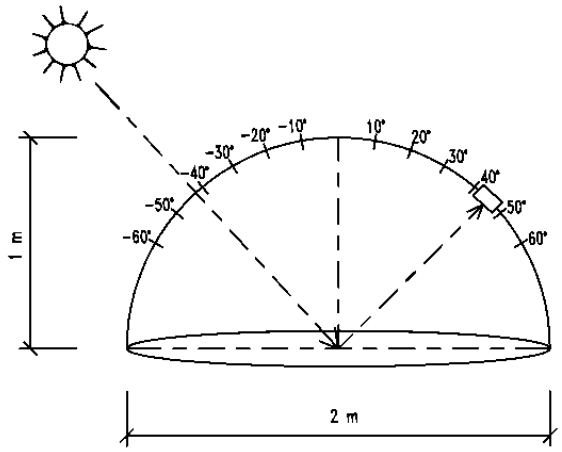


Fig. 2. Schematic of multiangular observation.

2015–2016. The selected wheat variety was Yanzhan 4110, which was susceptible to powdery mildew. The previous crop was corn, and the straw was crashed and returned to the field. The soil was sandy loam. Before sowing, the 0–30 cm soil contained 0.97 g·kg^{-1} of total nitrogen (N), 28.44 mg·kg^{-1} of available phosphorus (P), 114.68 mg·kg^{-1} of available potassium (K), and 10.2 g·kg^{-1} of organic matter. The amount of fertilizer applied was 270 kg·hm^{-2} pure N, 120 kg·hm^{-2} phosphorus pentoxide (P_2O_5), and 90 kg·hm^{-2} potassium oxide (K_2O). The experimental design had three replicates with plots with an area of 20.3 m^2 each. Sowing was carried out on October 12, 2015 and the cultivation management in each plot was consistent, adopted standard local practices for wheat production. In addition, Yanzhan 4110 were cultivated in pots in a greenhouse set aside for culturing diseases as sources of pathogens for the test inoculation. Three vaccination grades were set based on the frequency of inoculation, including severe (inoculation once a day, I3), moderate (inoculation once every two days, I2), and mild (inoculation once every three days, I1). A noninoculated control area (I0) was also set up. In the jointing stage, the powdery mildew pathogen was inoculated, and the artificial inoculation was continued until the lowest level of treatment began to translate into obvious disease symptoms. To create a suitable inoculation environment, the inoculated plot was covered with a transparent plastic shed (opened at 8:00–17:00), and the shed was maintained in a humid environment at a temperature of about 10 °C. Generally, after approximately ten days, white spores would emerge on the plant leaves, and once infection was obvious from the jointing to the booting stage, the transparent plastic shed would be removed. At the booting stage, the flowering stage, and the filling stage, the canopy spectra were collected and disease indices investigated by plant sampling (Fig. 1).

Experiment 2 (Exp.2): The inoculation test was conducted in a field at Henan Agricultural University Science and Education Demonstration Park (34° 51′ N, 113° 35′ E) in 2016–2017. The wheat variety was still Yanzhan 4110. The previous crop was corn, and the straw was crashed and returned to the field. Before sowing, the 0–30 cm soil contained 1.05 g·kg^{-1} of total N, 32.02 mg·kg^{-1} of available P, 118.32 mg·kg^{-1} of available K, and 11.7 g·kg^{-1} of organic matter. The amount of fertilizer applied was 270 kg·hm^{-2} pure N, 120 kg·hm^{-2} P_2O_5, and 90 kg·hm^{-2} K_2O. The experimental treatments had three replicates, each plot measuring 20.3 m^2. Sowing was conducted on October 18, 2016. Inoculation, plant sampling, and spectral data determination were performed as in experiment 1.

Experiment 3 (Exp.3): The experiment was conducted in a natural diseased field in 2016–2017. The experiment plot was located near a fence and a livestock farm in relatively low-lying terrain and the microclimate characteristics (poor ventilation, high humidity, and numerous sources of disease sources), which were conducive for the natural incidence of WPM and the disease was severe. The wheat varieties in the field were Yanzhan 4110, which was susceptible to powdery mildew, and Guomai 301, which was moderately susceptible to powdery mildew. The previous crop was corn, and the straw was crashed and returned to the field. Before sowing, the 0–30 cm soil contained 1.20 g·kg^{-1} of total N, 34.52 mg·kg^{-1} of available P, 120.12 mg·kg^{-1} of available K, and 15.7 g·kg^{-1} of organic matter. The amount of fertilizer applied contained 270 kg·hm^{-2} pure N, 120 kg·hm^{-2} P_2O_5, and 90 kg·hm^{-2} K_2O. Sowing was conducted on October 18, 2016. The experimental design had three replicates with plots with an area of 20.3 m^2 each. The data collection period was similar to the collection period in experiment 1.

B. Determination of Canopy Multiangle Spectral Data

The multiangle observation device used in this study was based on the Field Goniometer System and has a similar design to the University of Lethbridge Goniometer System (ULGS-I) [30], which is composed of a ring with a radius of 100 cm and a removable arch for affixing the sensor (Fig. 2). The spectrometer probe was placed at a distance of 100 cm from the target by adjusting four extendable legs at different growth stages. The multiangle reflectance data of wheat canopy was observed in the principal plane of the sun (SPP, 10° offset from the SPP and then assume symmetry to minimize the shadow cast by the arc). The observation angle perpendicular to the wheat canopy was defined as 0°, the observation direction was defined as the backward direction (−) when it was on the same side as the sun, and the observation direction was defined as the forward direction (+) when it was on the opposite side of the sun. The observation direction was

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from the backward to the forward, including 13 observation angles (±60°, ±50°, ±40°, ±30°, ±20°, ±10°, and 0°). The reflectance spectra were collected using a FieldSpec Handheld Spectrometer (Analytical Spectral Devices, Boulder, CO, USA), with a spectral range of 325–1075 nm, and a high spectral sampling interval and a spectral resolution of 1.5 and 3.5 nm, respectively. Spectral measurements were conducted with the bare fiber optic cable which has an angular field of view (FOV) of 25°. The normal operating procedure for these measurements is to place the focus inside the canopy. With the VZA increasing, the major axis of the FOV ellipse dramatically changed, which resulted in a top-of-the-canopy footprint of 0.44 m in diameter at nadir and 1.05 m (end to end) at 60° VZA, and covered at least three rows of wheat (row spacing = 0.18 m). Spectral information was collected at 10:00–14:00 local time under sunny, cloudless, and windless weather conditions. Each viewing angle was measured five times and the average value of the output obtained as the spectral reflectance of the point. Thirteen viewing angle data were acquired from each plot within 5 min, which ensured minimal change in solar illumination angle. The reflection spectra were normalized and optimized with a reference 40 × 40 cm^2 barium sulfate plate before spectral data acquisition at each monitoring point.

C. Disease Severity Determination

At each spectral collection point, 20 wheat plants were randomly selected to record the incidence of WPM. To minimize human error, the same plant protection professionals oversaw all disease investigations. Severity at a sampling point was expressed based on the disease severity in a fully expanded leaf, and the severity of powdery mildew was calculated based on the grid method (plaque area as a percentage of the fully expanded leaf). The severity of conventional powdery mildew was divided into nine grades, including 0%, 1%, 10%, 20%, 30%, 45%, 60%, 80%, and 100%. The number of wheat leaves under each severity level was recorded separately, and then the following formula was used to calculate the disease index (DI):

$$\mathrm{DI} = \frac{\sum xf}{n \sum f} \times 100 \tag{1}$$

where x is the level of each gradient, n is the highest disease gradient ($n = 9$), and f is the total leaf number for each gradient.

To eliminate the influence of planting density, tillering ability, and growth difference on the disease severity calculations, this study applied the modified disease index (mDI) proposed by Feng *et al.* [9] to characterize the incidence of WPM and to carry out analyses of related calculation

$$\mathrm{mDI} = \frac{\mathrm{DI}}{\mathrm{LAI}}. \tag{2}$$

D. Data Analysis and Application

Spectral data at each observation angle were analyzed as independent data sets, and spectral reflectance data were read at intervals of 2 nm in the 400–900 nm range. In a matrix arrangement mode, the spectral parameter (X) for all possible two-band combinations was fit to the WPM disease severity index (Y), the corresponding coefficient of determination (R^2) was calculated, and the concentration R^2 contour map was plotted. The two-band combination method applied in this study included the Normalized Difference Vegetation Index (NDVI) and the ratio vegetation index (RVI). The contour map plotted in this study was implemented using self-programming in the MATLAB 7.0 platform (The Math Works, 2000).

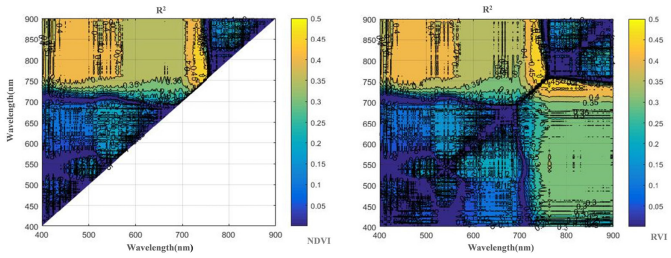


Fig. 3. Contour map of coefficients of determination for the relationships between canopy mDI and NDVI and RVI.

The development of new parameters was based on the disease identification spectral indices [Healthy Index (HI), Cercospora Leaf Spot Index (CLS), Sugar Beet Rust Index (SBRI), and PMI] developed by Mahlein *et al.* [3] and the LAI monitoring spectral index [Normalized Difference Spectral Index (NDSI)] developed by Cao *et al.* [31]. A novel three-band spectral parameter was developed based on (5)–(8)

$$\mathrm{NDVI}(\lambda 1, \lambda 2) = (R_{\lambda 1} - R_{\lambda 2})/(R_{\lambda 1} + R_{\lambda 2}) \tag{3}$$

$$\mathrm{RVI}(\lambda 1, \lambda 2) = R_{\lambda 1}/R_{\lambda 2} \tag{4}$$

$$\mathrm{mNDVI}_1(\lambda 1, \lambda 2, \lambda 3) = (R_{\lambda 1} - R_{\lambda 2} \pm k1 \times R_{\lambda 3}) / (R_{\lambda 1} + R_{\lambda 2} \pm k2 \times R_{\lambda 3}) \tag{5}$$

$$\mathrm{mNDVI}_2(\lambda 1, \lambda 2, \lambda 3) = (R_{\lambda 1} - R_{\lambda 2})/(R_{\lambda 1} + R_{\lambda 2}) \pm k1 \times R_{\lambda 3} \tag{6}$$

$$\mathrm{mRVI}_1(\lambda 1, \lambda 2, \lambda 3) = (R_{\lambda 1} \pm k1 \times R_{\lambda 3})/(R_{\lambda 2} \pm k2 \times R_{\lambda 3}) \tag{7}$$

$$\mathrm{mRVI}_2(\lambda 1, \lambda 2, \lambda 3) = R_{\lambda 1}/R_{\lambda 2} \pm k3 \times R_{\lambda 3}. \tag{8}$$

In the formula, $\lambda 1$, $\lambda 2$, and $\lambda 3$ are wavelengths (nm) and $\lambda 1 \neq \lambda 2 \neq \lambda 3$ (400–900 nm); $R_{\lambda 1}$, $R_{\lambda 2}$, and $R_{\lambda 3}$ are reflectance corresponding to wavelengths $\lambda 1$, $\lambda 2$, and $\lambda 3$, respectively. K ($k1$, $k2$, and $k3$) is the coefficient of the third band, where k is in the −1, −0.5, 0, 0.5, and 1 range.

PMI has been developed in previous studies and the conventional hyperspectral parameters that correlated well with the powdery mildew severity are listed in Table I.

III. Results

A. Relationship Between Two-Band Parameters and Disease Severity

The relationship between the two-band normalization and the ratio parameters of the original spectrum and the WPM severity based on the vertical observation angle were systematically studied (Fig. 3). According to the results, the sensitive areas of the severity were mainly located in the red and

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TABLE I
SUMMARY OF SELECTED SPECTRAL INDICES REPORTED IN THE LITERATURE

Spectral index	Formulation	Reference
Powdery mildew index	PMI= (R_{520}-R_{584})/(R_{520}+R_{584})+R_{724}	Mahlein et al.(2013)[3]
Red edge reflectance-ratio indices (Vogelmann)	RERIvog= (R_{734}-R_{747})/(R_{715}+R_{726})	Zarco-Tejada et al.(2001)[32]
Pigment specific 1position1d difference	PSNDc= (R_{800}-R_{470})/(R_{800}+R_{470})	Blackbum(1998)[33]
Red edge 1position with inverted gauss	$R(\lambda)= R_S-(R_S-R_0)\exp(\frac{-(\lambda 0-\lambda)^2}{2\delta^2})$	Miller et al.(1990)[34]
Ratio index of the double-peak area	$RIDA=\int_{718}^{755}\frac{dR\lambda}{d\lambda}d\lambda/\int_{680}^{718}\frac{dR\lambda}{d\lambda}d\lambda$	Feng et al.(2014)[35]
Normalized difference index of the double-peak areas	$NDDA=\left(\int_{718}^{755}\frac{dR\lambda}{d\lambda}d\lambda/\int_{680}^{718}\frac{dR\lambda}{d\lambda}d\lambda\right)/\int_{680}^{755}\frac{dR\lambda}{d\lambda}d\lambda$	Feng et al.(2014)[35]
Soil-adjusted vegetation index	SAVI=(1-0.08)×(R_{825}-R_{735}) / (R_{825}+ R_{735}-0.08)	Huete (1988)[36]
Red edge chlorophyll index-1	CI red-edge1=[($R_{750\text{-}800}$)/($R_{695\text{-}740}$)]-1	Gitelson et al.(2003)[37]
Red edge chlorophyll index-2	CI red-edge2=[($R_{840\text{-}870}$)/($R_{720\text{-}730}$)]-1	Gitelson et al.(2005)[38]
Red edge chlorophyll index-3	Cired-edge3= $R_{790}/R_{720}-1$	Gitelson et al.(2005)[38]
Normalized difference red edge	NDRE= (R_{790}-R_{720})/(R_{790}+R_{720})	Barnes et al.(2000)[39]
Red edge 1position with linear four-point interpolation	REPlfi=700+4×[(R_{670}+R_{780})/2-R_{700}]/(R_{740}-R_{700})	Guyot et al. (1988)[40]
Red edge reflectance curve area	REFCA=Sum(R_{680},R_{780})/101×R_{780}	Ren et al.(2011)[41]
Red edge symmetry	RES= (R_{718}-R_{675})/(R_{755}-R_{675})	Ju et al.(2010)[42]
Normalized powdery mildew index	NPMI=(R_{744}-R_{762})/(R_{744}+R_{762}-R_{721})	This study
Ratio powdery mildew index	RPMI=R_{744}/R_{762}-0.5×R_{710}	This study

near-infrared regions ($R^2 > 0.5$). The normalized difference sensitive bands were concentrated at $\lambda 1$ (720–770 nm) and $\lambda 2$ (720–870 nm), and the RVI sensitive bands were concentrated at $\lambda 1$ (710–760 nm) and $\lambda 2$ (720–900 nm). By comparison, the optimal band combinations for the two types of parameters were mainly concentrated at 736–750 and 746–766 nm, respectively. Considering the optimal spectral parameters as the representative parameters, a scatterplot was drawn using a regression equation to visualize the equation fitting effect (Fig. 4). The index with the combination of $\lambda 1 = 744$ nm and $\lambda 2 = 762$ nm had the highest accuracy and its root-mean-square error (RMSE) was the lowest (NDVI: $R^2 = 0.547$, RMSE = 3.078; RVI: $R^2 = 0.551$, RMSE = 3.067).

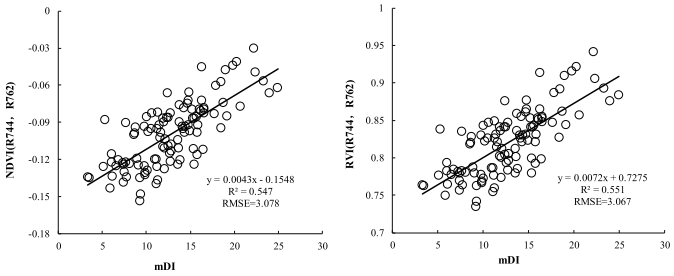


Fig. 4. Quantitative relationships between NDVI (R_{744}, R_{762}) and RVI (R_{744}, R_{762}) and winter wheat canopy powdery mildew mDI.

B. Relationship Between NPMI, RPMI, and Disease Severity

To further improve the monitoring accuracy of disease severity, we introduced a third band, λ_3, and its coefficient value, K, to optimize the above two-band red-edge parameters, NDVI (R_{744}, R_{762}) and RVI (R_{744}, R_{762}). The $mNDVI_1$, $mNDVI_2$, $mRVI_1$, and $mRVI_2$ forms have been verified, and their optimized forms are $mNDVI = (R_{744} - R_{762} + K_1 \times \lambda_3)/(R_{744} - R_{762} + K_2 \times \lambda_3)$ and $mRVI = R_{744}/R_{762} + K_3 \times \lambda_3$. Based on calculation, the K values were determined to be -1, -0.5, 0, 0.5, and 1; and the λ_3 range was 400–900 nm. In addition, in the mNDVI form, when K_1, K_2, and λ_3 were 0, -1, and 721 nm, respectively, the optimal NDVI form was $(R_{744} - R_{762})/(R_{744} + R_{762} - R_{721})$ [Normalized Powdery Mildew Index (NPMI)], which had the greatest correlation with disease severity, and the correlation coefficient reaches

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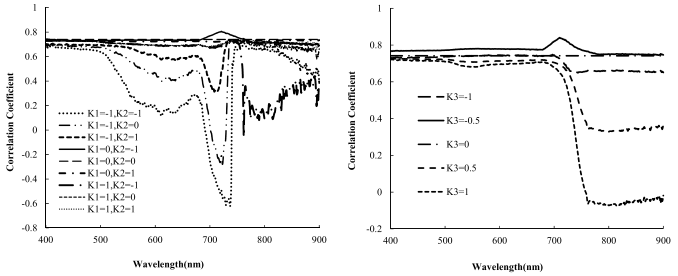


Fig. 5. Correlation between mNDVI $((R_{744} - R_{762} + K_1 \times \lambda_3)/(R_{744} - R_{762} + K_2 \times \lambda_3))$ and mRVI $(R_{744}/R_{762} + K_3 \times \lambda_3)$ and disease severity. $k1$, $k2$, and $k3$ are the coefficients of the third band, k is in the -1, -0.5, 0, 0.5, and 1 range. $\lambda 3$ is the third band.

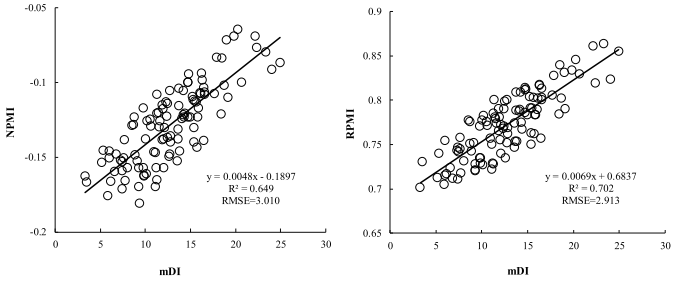


Fig. 6. Quantitative relationships between NPMI and RPMI, and mDI.

0.806. In the mRVI form, when K_3 and λ_3 were 0, -0.5, and 710 nm, the optimal RVI form was $R_{744}/R_{762} - 0.5 \times R_{710}$ [Ratio Powdery Mildew Index (RPMI)], which had the greatest correlation with disease severity, and the correlation coefficient could reach 0.840 (Fig. 5). Based on the unified regression analysis of two-year-data, the optimal coefficients between NPMI and RPMI, and disease severity, reached 0.649 and 0.702, respectively (Fig. 6). The results indicated that the fitting accuracy of NPMI and RPMI against mDI was higher than in the two-band constructed parameters, NDVI (R_{744}, R_{762}) and RVI (R_{744}, R_{762}), against mDI, and the R^2 increased by 18.65% and 27.40%, respectively, while RMSE decreased by 2.21% and 5.02%, respectively.

C. Relationship Between Disease Severity and Spectral Parameters Under Different Observation Angles

The polar plots for severe infected wheat were inspected across six representative bands (Fig. 7). The canopy reflectance in the backscattering direction was higher than in the forward-scattering direction in SPP and approximately symmetrical in perpendicular plane. Based on the correlation changes between the same spectral parameters at different observation angles and the correlation changes between different spectral parameters at the same observation angles, the relationship between the selected spectral parameters and disease severity was further analyzed. Table II shows that the correlation between the selected hyperspectral parameters and the two novel parameters (NPMI and RPMI) and mDI exhibited similar angular effects. The correlation between each parameter and mDI was better in the forward observation direction than in the backward observation direction, and the correlation

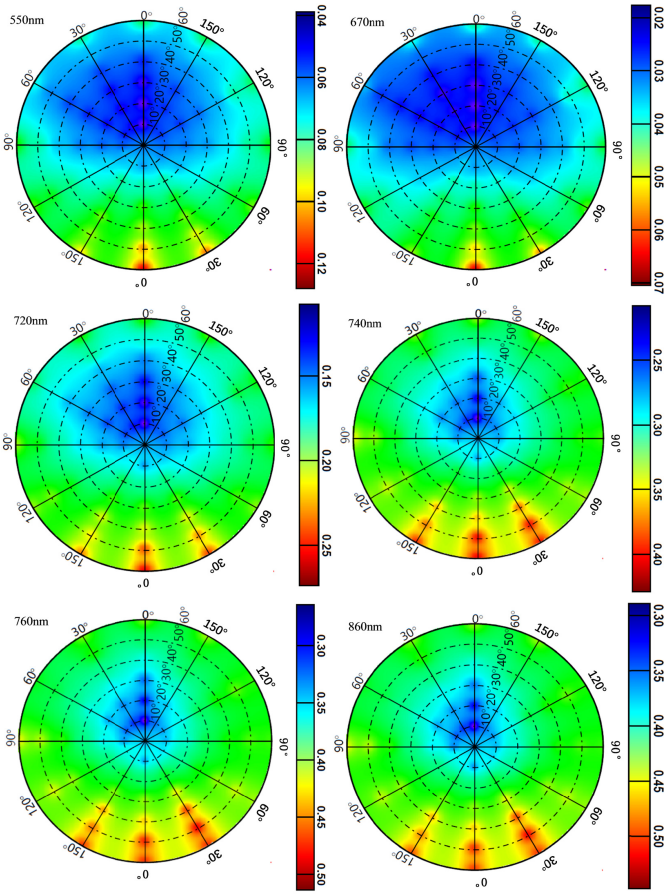


Fig. 7. Bidirectional Reflectance Distribution Function polar plots for severe infected wheat in the 550, 670, 720, 740, 760, and 860 nm of the spectrum reflectance. (DI = 76.39%, solar zenith angle is near 40°, and the plane of illumination is vertical on the page).

between the spectral parameters and mDI gradually increased from $-60°$ to $+30°$. The correlation decreased gradually with an increase in backward observation angle. The correlation between Ratio Index of the Double-peak Area (RIDA) and mDI exhibited superior angular stability from $-30°$ to $+30°$, while the correlation between other spectral parameters and mDI was relatively stable from $+10$ to $+40°$. The corresponding optimal observation angles of different spectral parameters were different and were mainly concentrated at $+10°$ to $+30°$ and mostly in $+10°$ as a whole. There were also significant differences in the performance of spectral parameters. The hyperspectral parameters that were positively correlated with mDI under different observation angles were red edge reflectance-ratio indices (Vogelmann) (RERIvog), RIDA, normalized difference index of the double-peak areas (NDDA), red edge reflectance curve area (REFCA), red edge symmetry (RES), NPMI, and RPMI, although they were negatively correlated with the remaining parameters. Compared with conventional spectral parameters, the novel parameters NPMI and RPMI had a higher correlation with mDI, and RPMI had a higher correlation than NPMI. Both spectral parameters exhibited the highest correlations at $+10°$, at 0.826 and 0.852, respectively. As a key index for the identification and classification of powdery mildew, PMI exhibited the least sensitivity to disease

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TABLE II
CORRELATION BETWEEN CANOPY HYPERSPECTRAL PARAMETERS AND WPM SEVERITY AT DIFFERENT OBSERVATION ANGLES

VIs	-60°	-50°	-40°	-30°	-20°	-10°	0°	+10°	+20°	+30°	+40°	+50°	+60°
PMI	-0.130	-0.129	-0.149	-0.184	-0.208	-0.243	-0.266	-0.259	-0.289	-0.289	-0.277	-0.278	-0.233
RERIvog	0.442	0.526	0.571	0.599	0.623	0.651	0.651	0.672	0.671	0.664	0.648	0.651	0.619
PSNDc	-0.351	-0.499	-0.469	-0.535	-0.633	-0.633	-0.661	-0.674	-0.649	-0.648	-0.602	-0.604	-0.528
REPig	-0.478	-0.548	-0.578	-0.598	-0.621	-0.644	-0.647	-0.663	-0.654	-0.666	-0.643	-0.634	-0.595
RIDA	0.553	0.612	0.649	0.679	0.680	0.684	0.694	0.700	0.699	0.692	0.665	0.666	0.631
NDDA	0.512	0.565	0.597	0.622	0.634	0.661	0.660	0.675	0.669	0.682	0.650	0.648	0.614
SAVI	-0.490	-0.573	-0.600	-0.630	-0.667	-0.686	-0.693	-0.708	-0.695	-0.696	-0.675	-0.669	-0.639
CI re1	-0.430	-0.524	-0.567	-0.603	-0.627	-0.652	-0.654	-0.673	-0.667	-0.673	-0.648	-0.646	-0.610
CI re2	-0.438	-0.534	-0.572	-0.605	-0.632	-0.659	-0.66	-0.678	-0.671	-0.677	-0.657	-0.652	-0.621
CI re3	-0.430	-0.524	-0.565	-0.602	-0.617	-0.647	-0.650	-0.663	-0.668	-0.672	-0.649	-0.645	-0.618
NDRE	-0.470	-0.55	-0.580	-0.617	-0.637	-0.664	-0.663	-0.678	-0.671	-0.682	-0.65	-0.646	-0.612
REPlfi	-0.518	-0.59	-0.616	-0.636	-0.654	-0.678	-0.682	-0.695	-0.687	-0.694	-0.673	-0.671	-0.638
REFCA	0.473	0.554	0.568	0.61	0.636	0.652	0.664	0.653	0.657	0.67	0.637	0.637	0.598
RES	0.517	0.563	0.598	0.617	0.634	0.658	0.66	0.672	0.669	0.676	0.645	0.643	0.614
NPMI	0.615	0.659	0.694	0.708	0.756	0.785	0.806	0.826	0.807	0.765	0.755	0.723	0.666
RPMI	0.612	0.658	0.685	0.707	0.77	0.804	0.838	0.852	0.847	0.821	0.785	0.725	0.671

severity, and the correlations under different observation angles were less than 0.29.

D. Influence of Observation Angle Range on Disease Severity Monitoring Model

To further explore the optimal ranges of observation angles, four conventional parameters [Normalized Difference Red Edge (NDRE), Soil Adjusted Vegetation Index (SAVI), RIDA, Red Edge Position with Linear Four-point Interpolation (REPlfi)] with good correlation and two novel parameters (NPMI and RPMI) were selected to compare the ability to monitor the disease severity under different observation angle ranges. As illustrated in Fig. 8, the correlation between the six selected parameters and the mDI in the forward observation direction (+60° to +0°, $R^2 = 0.42$–0.61, RMSE = 2.77–3.40) was higher than in the backward observation direction (−60°–0°, $R^2 = 0.35$–0.5, RMSE = 2.93–3.73). The correlations between parameters and mDI were optimal at +10° to +20° ($R^2 = 0.46$–0.72, RMSE = 2.35–3.30). The second most preferred angle range was +10° to +30°, with R^2 and RMSE at 0.45–0.71 and 2.42–3.31, respectively. NPMI and RPMI performed well in the six parameters. The monitoring accuracy was 0.5–0.72 and 0.48–0.67 under different observation angle ranges, and the angle ranges with high monitoring accuracy were concentrated at 0° to +30°, +10° to +30°, +10° to +20°, with R^2 and RMSE of RPMI and NPMI in the 0.70–0.72, 2.35–2.43, 0.64–0.67, and 2.58–2.68 ranges, respectively.

E. Establishment of Monitoring Model

The fitting ability of the model was further demonstrated by selecting the novel parameters, NPMI and RPMI, which exhibited a good relationship with mDI, and a regression analysis was carried out by adopting an appropriate suitable forward observation angle (0° to +30°) as an example. Fig. 9 shows that the monitoring accuracy and RMSE of NPMI and RMPI for mDI were 0.638 and 2.68, and 0.704 and 2.43, respectively. Comparing the two spectral parameters, the fitting accuracy of the RPMI model was 10.34% higher than that of NPMI, and the RMSE decreased by 9.32%. Therefore, the novel index RPMI could be used to monitor the disease status of winter WPM in the 0° to +30° range.

IV. DISCUSSION

A. Sensitive Bands and VIs for Disease Monitoring

The transition region between the strong absorption of chlorophyll in the red band and the high-reflection platform

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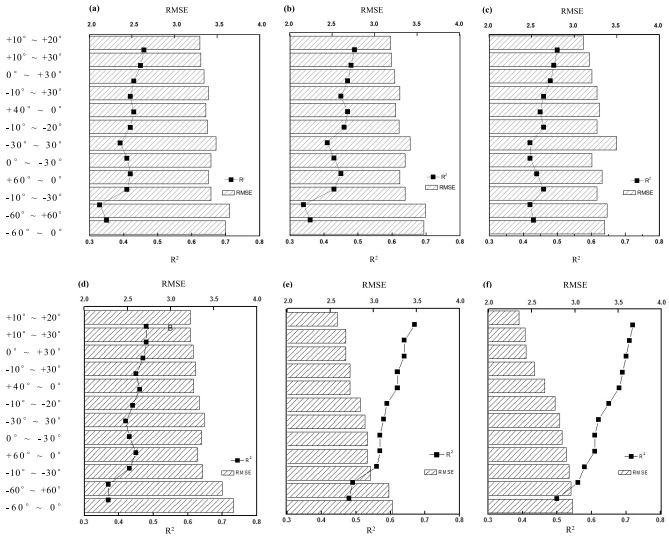


Fig. 8. Comparison of the relationships between different observation angle ranges and disease severity. (a) NDRE. (b) SAVI. (c) RIDA. (d) REPLfi. (e) NPMI. (f) RPMI.

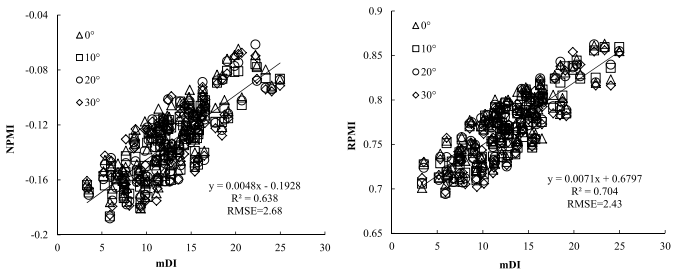


Fig. 9. Quantitative relationships between vegetation index and mDI in the 0° to +30° range.

formed by multiple scattering in the near-infrared region is referred to as the red edge (680–780 nm) [43]. The red edge is a key region in research on the spectral characteristic curve. Numerous studies have applied the red-edge band, the red-edge position, and the red-edge parameter to assess crop physiological and biochemical parameters [1], [3], [31]. *Blumeria graminis* is fungal pathogen that frequently causes disease in wheat fields through soil and airborne transmission. Infection by the pathogens could damage chlorophyll in wheat plant, damage the internal structures of leaves, and alter canopy structure. When plants are damaged by diseases and insect pests, the red edge shifts to the short wave direction and the-red edge amplitude decreases [44]. Such typical spectral characteristics are different from those observed in healthy crops. The preliminary work of our laboratory also pointed out that the sensitive band of WPM was concentrated at 690–718 nm at the canopy scale [6]. In addition, Cao *et al.* [8] pointed out that red-edge parameters (such as red-edge slope and red-edge area) were suitable for monitoring WPM. The results of this study revealed that the sensitive bands for WPM were concentrated in the red-edge region (710–870 nm), and the NDVI (744,762) and RVI (744,762) formed by the red-edge bands at 744 and 762 nm in the sensitive area were highly correlated with mDI. The third band introduced based on the two bands was mainly concentrated in the red edge, which demonstrated the importance of the red-edge band for the monitoring of WPM.

Spectral bands and parameters that are sensitive to disease will vary from crop to crop [3]. The PMI developed by the previous researchers was less effective in monitoring disease severity in this study. This may be because the parameter was developed when monitoring beet powdery mildew, and two bands of the PMI were green bands. This study observed that the sensitive bands of WPM were concentrated in the red edge. The novel spectral parameters, NPMI and RPMI, which were developed based on three red-edge bands, were highly accurate in monitoring powdery mildew. In addition, the high-performing traditional VIs (RERIvog, red edge chlorophyll index1 (CIre1), CIre2, CIre3, Replfi, and REFCA) were also composed mainly of red-edge bands and the results were stable and consistent. Certainly, in addition to the red-edge band, the green band is sensitive when applied in the monitoring diseases [43]–[45]. Our preliminary study [9] applied the two green bands to develop the dual-green vegetation index to enhance the WPM monitoring accuracy, and the results were inconsistent with the results of the spectral parameters developed in this study. A preliminary study in our laboratory screened sensitive bands based on the reflectance ratio of the healthy population and the diseased population. The DI can be monitored using only two narrowband reflectance. In this study, the screening of the powdery mildew sensitive band only needs to determine the proportion of the vegetation population infected by the pathogen; however, the DI has to be monitored using three narrowband reflectance. Therefore, the data processing methods are different, and the sensitive bands and spectral parameters obtained by screening for monitoring diseases are also inconsistent.

B. Sensitive Angles of Disease Monitoring

Numerous studies have shown that canopy spectral reflectance is influenced by factors such as canopy structure, soil background, light intensity, and direction of observation. The observation angle influences canopy reflectance, which in turn affects VI performance [46]. In recent years, some studies have proposed that the VZA influences the relationship between spectra and biochemical characteristics of vegetation. In addition, the illumination intensity and shadow area would affect the performance of spectral parameters and the backward observation direction was more appropriate for the monitoring of crop physiological indicators [47]. Moreover, according to Oppelt [48], the $-36°$ and $-55°$ in the backward observation had the optimal inversions for chlorophyll a and chlorophyll b in the wheat light layer. The results of the preliminary study conducted in our laboratory also suggested that the backward observation angle could improve the monitoring accuracy of LNC in wheat and that the spectral reflectance at an observation angle of $-20°$ was the most sensitive to LNC [28]. The sensitive angles of growth indicators of crops under abiotic stress growth conditions are mainly in the backward direction, and the corresponding sensitive angles of different physiological indicators are different. When a

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disease occurs, the plant lesions accelerate plant senescence, morphological change, structural damage, and internal physiological metabolism impaired, which are accompanied by a series of complex changes associated with a disease. Such changes would considerably influence the response of the canopy reflectance spectrum. Therefore, under disease stress, the appropriate angle of monitoring plant morphological and physiological indicators could be considerably different from the appropriate observation angle under abiotic stress. Oberti *et al.* [49] recommended $+40°$ to $+60°$ as a sensitive angle range for monitoring powdery mildew in grape leaves in the early and mid-term using multispectral imaging methods. For the first time, this study systematically analyzed the capacity of traditional VIs and novel spectral parameters to monitor the mDI of WPM under different observation angles. According to the results, the forward observation angle was more appropriate for monitoring the mDI, which was optimal at $+10°$, compared with the backward observation angle. This finding under disease stress is inconsistent with the results of LNC monitoring under N stress in the previous study [28]. The sensitive angles of powdery mildew monitoring identified in this study were significantly different from the sensitive observation angles identified for N and chlorophyll under abiotic stress. The difference could be because the unique characteristics of powdery mildew influence the distribution of canopy light radiation so that the difference between the reflectance of light on a plaque and the reflectance of light on healthy leaves is more significant. The white plaque reflects less light in the forward direction with more shadows, in turn reflecting the disease information more effectively. The conclusion also confirms that different observation angles should be selected while considering the observation target to enhance monitoring accuracy when using multiangle remote sensing.

In practice, detection of disease at low severity is more meaningful than at medium and high severity, which may be beneficial to prevention efforts. Some studies have focused on the early detection of plant diseases using different algorithms. It has been reported that the distance change between the red-edge position and the yellow edge position could be applied in the identification of wheat stripe rust in the early stages of an epidemic [44]. In addition, FCI could achieve the early detection of *Fusarium* head blight [10]. Support vector machine (SVM) based on hyperspectral reflectance could be used to distinguish Cercospora leaf spot, leaf rust, and powdery mildew in sugar beet leaves early [50]. Furthermore, combined SVM and hyperspectral imaging have been evaluated for their effectiveness in the early detection of pear fruit disease [51]. A neural network has also been developed for the early detection of laurel wilt disease in avocado [52], in addition to septoria leaf blotch and yellow rust [53]. A study revealed that stepwise discriminant analysis was the optimal classifier for the detection strawberry Anthracnose crown rot at early stages [54]. These algorithms could achieve satisfactory monitoring accuracy, but the relationships between the disease characteristics and spectrum are seldom considered. Due to the bottom-up spatial characteristics of powdery mildew, the early detection of powdery mildew can be achieved by collecting bottom growth information of winter wheat. Multiangular remote sensing could expand the detection capacity to the lower canopy levels [55], which demonstrates that multiangle remote sensing technology has great potential and advantages in the early identification of powdery mildew symptoms and in high precision monitoring. In this study, the severity of powdery mildew was divided into nine grades, including low, medium, and high severity levels. The performances of traditional and novel VIs were compared under 13 VZAs. The optimal angle was $+10°$, and the smaller observation angles near the vertical angle mainly reflected information from the middle and lower crop layers (early WPM disease symptoms). The findings indicated that $+10°$ not only improved the monitoring accuracy of medium and high severity disease levels but also enhanced the early detection capacity when the crop was in the initial WPM infection stages. The findings also demonstrated multiangular remote sensing as an effective method for improving the monitoring accuracy or early detection of WPM. The results of this study could facilitate the early prevention and high precision control of wheat diseases, reducing pesticide dosages, improving prevention efficiency, and protecting the environment.

C. Appropriate Angle Ranges for Disease Monitoring

A narrow observation angle range has certain limitations in actual production operation environments. Selecting or developing a sensitive VI and a suitable wide-angle range to develop a more comprehensive monitoring model could improve the WPM monitoring capacity of remote sensing technology in actual production operations. Therefore, we require appropriate disease indices that are not sensitive to VZAs in under multiangular remote sensing, which is why the novel parameter in this study was developed. According to Waltershea *et al.* [56], smaller observation angles are more appropriate for monitoring physiological indicators of crops compared with vertical and large angles. Conversely, larger observation angles are more suitable for monitoring crop upper LAI information [57]. Multiangle remote sensing satellites were applied to monitor forest LAI and chlorophyll indicators and the results showed that larger observation angles were more suitable for the monitoring of physiological and biochemical indicators of narrow-waveband VIs [23]. The angle ranges screened by previous scholars when using multiangle remote sensing to monitor target parameters vary. A small angle near the vertical angle was suitable for monitoring the growth information of the middle and lower layers of wheat [28]. In addition, multiangle remote sensing was used to monitor chlorophyll concentration in the wheat canopy and concluded that $\pm 20°$ and $\pm 40°$ were more suitable for monitoring chlorophyll concentrations in the middle layers of wheat canopy [25]. The correlations among parameters in this study decreased with an increase in the observation angle. The optimal observation angle was mostly $+10°$; however, a stable monitoring accuracy was also observed in the $+10°$ to $+30°$ range, which indicated that the wide range of observation angles was still suitable for monitoring WPM, which was similar to the results of a previous study on LNC [28]. The $+10°$ to $+30°$ angle range identified in this study was a suitable angle range for

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powdery mildew monitoring, which is consistent with the bottom-up epidemiological characteristics of WPM. Based on the insensitivity of the red edge to the change in viewing angle, the novel three-band red-edge index is proposed. The K value, as the adjustment coefficient in this parameter, should vary based on different disease types and test environments. In this article, the K value is 0.5, which could well coordinate the relationship between the three red-edge bands, so that RPMI could better capture the powdery mildew occurrence and development data. The red-edge band adopted in this article is extracted from hyperspectral narrowband information. Therefore, it required a higher spectral resolution. If the parameter and model are extrapolated to a wide-band sensor, the monitoring accuracy may decrease due to overlap of band information. With the advancement of hyperspectral filter technologies, the parameter and its monitoring method have great application prospects and value in the prevention and precise control of vegetation diseases in the future. Compared with traditional VIs, the novel parameter, RPMI, addresses the shortcomings of vertical observation angles and expands the narrow sensitivity angle range to wide angle ranges ($+10°$ to $+30°$), which potentially increases the flexibility and operability of remote sensing monitoring. In addition, with the development of unmanned aerial vehicle (UAV) technologies, RPMI provides a reference for UAV to consider flight paths and observation directions when monitoring wheat diseases and recommends a range of angles for the monitoring of WPM using multiangle remote sensing satellites.

V. Conclusion

The pathogenesis of WPM is from the plant bottom layer to the plant upper layer, and multiangle remote sensing has a key advantage in the monitoring of the mDI of WPM in a timely manner. In this study, RPMI $[R_{744}/R_{762} - 0.5 \times R_{710}]$ was developed based on the red-edge band sensitive to powdery mildew. The forward observation direction was more suitable for monitoring WPM than the backward observation direction in the SPP, especially $+10°$ was the optimal observation angle. Considering the feasibility and convenience in actual production operation environments, the observation angle of RPMI model could be extended to the $0°$ to $+30°$ range. The results of this study are beneficial to the early monitoring of WPM through acquiring more information about plant middle and lower layers, so as to early prevent WPM and improve the control efficiency. Meanwhile, it has a significant contribution to a growing body of knowledge on the topic of angular remote sensing, and potential applications that could offer option fly direction based on UAV or multiangular remote sensing satellite.

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